

Using Time Series Data to Predict Power Outage Duration

We KANT, but we can

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Goal Statement: Be able to predict the duration of power outages based on factors about the event affecting power across the United States.

Research Question: Can we develop a model that can predict the duration of U.S. power outages caused by various events with a mean value error of less than 30 minutes?

Narrative:

In an age where electricity has become a necessity in our day to day lives, power outages bring an unwarranted and sometimes highly detrimental problem to individuals and businesses. Accurately being able to predict how long it will take to recover the power is crucial for efficiency in academic, personal, and corporate environments. Our personal motivation to pursue this work lies in the fact that some of our group members have lived in apartment complexes that have lost power and the unease of not knowing when the power will come back on can be difficult to manage especially when as students there are still due dates and deadlines to meet.

We will use a dataset that includes recorded power outages across the United States. We want to create a model that can predict the length of power outages based on past records. As we create our model for prediction capacities we can compare it against general time expectations to determine if power is being restored at adequate speeds. Minor electrical problems should take 1-2 hours, residential areas 2-4 hours and major commercial or industrial outages around 1-3 hours to fix [1].

Modeling Approach:

For our modeling approach we will compare three different models to see which can best forecast the duration of power outages in the U.S. [3]. We will start with a linear regression model, which will serve as our baseline model. We will then run a random forest model and gradient boosting model as the other two models in our comparison. We will then compare the models using various evaluation metrics. Our main evaluation metric will be the mean value error (MAE), but we will also look at the R^2 score and the root mean squared error (RMSE).

References

- [1] "Emergency electrician response times: For different situations," ShockFree Electrical, <https://shockfreeelectrical.com/blog/emergency-electrician-response-times-for-different-situations/> (accessed Feb. 25, 2026).

[2] “Data.gov,” *Data.gov*, 2025.

<https://catalog.data.gov/dataset/event-correlated-outage-dataset-in-america> (accessed Feb 25, 2026).

[3] M. Mohan, T. Ayyalasomayajula, Santhosh Bussa, and S. Ayyalasomayajula, “Forecasting Home Prices Employing Machine Learning Algorithms: XGBoost, Random Forest, and Linear Regression,” *Journal of Engineering & Technology Advancements*, vol. 1, no. 1, pp. 2583–2646, doi: <https://doi.org/10.56472/25832646/JETA-V1I1P114>.